

Chapter 1: Complex Analysis

Mathematical Physics

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- 1 Taylor series representation
- 2 Complex variables
- 3 Derivatives
- 4 Path integrals
- 5 Taylor and Laurent series
- 6 Singularities
- 7 Residues
- 8 Residue calculus
- 9 Method of steepest descent

Where we are going

To analyze the special functions of physics we first build the tools of *complex analysis*.

- **Taylor and Laurent series** represent a function near a point, even near a singularity.
- **Analytic functions** (complex differentiable) obey the *Cauchy–Riemann equations* and admit power-series expansions.
- **Cauchy’s integral theorem** and the **residue theorem** turn contour integrals into a sum of local residues—the tool used later for Gamma/Beta identities and real integrals.
- A short catalog of **contour templates** (semicircle, keyhole, Jordan) shows how orientation and vanishing-arc estimates convert complex residues into real answers.
- The **method of steepest descent** extracts large-parameter asymptotics of contour integrals. Its real special cases (Laplace method, stationary phase) reappear for Stirling (Ch. 2) and for large-argument Bessel expansions (Ch. 5).

Matching a polynomial to a function

A Taylor polynomial replaces a function near one point by a polynomial with the same value, slope, curvature, and higher derivatives there.

Start at $x = 0$ and write the first few terms rather than a general sum:

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_Nx^N \quad (N \geq 3).$$

Differentiate and then set $x = 0$:

$$p(0) = a_0,$$

$$p'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots,$$

$$p'(0) = a_1,$$

$$p''(x) = 2a_2 + 6a_3x + \cdots,$$

$$p''(0) = 2!a_2,$$

$$p^{(3)}(x) = 6a_3 + \cdots,$$

$$p^{(3)}(0) = 3!a_3.$$

To match f , choose $a_0 = f(0)$, $a_1 = f'(0)$, $a_2 = f''(0)/2!$, and so on. The pattern is

$$a_k = \frac{f^{(k)}(0)}{k!}. \quad (1.1.1)$$

The factorial appears because differentiating x^k exactly k times gives $k!$.

Matching a polynomial to a function (cont.)

At a different center x_0 , use powers of the distance $x - x_0$:

$$T_N(x; x_0) = \sum_{k=0}^N \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (1.1.2)$$

For example, the quadratic approximation is

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2.$$

How accurate is it? For a real-valued f and real $x \neq x_0$, assume f has $N + 1$ continuous derivatives between x_0 and x , Taylor's theorem gives

$$f(x) = T_N(x; x_0) + R_N(x), \quad R_N(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} (x - x_0)^{N+1}, \quad (1.1.3)$$

where ξ lies between x_0 and x . At $x = x_0$ the error is zero. Thus a bound on the next derivative gives a bound on the error.

When $R_N(x) \rightarrow 0$, the polynomial becomes an exact infinite series:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (1.1.4)$$

Having derivatives of every order is not, by itself, enough to guarantee this last equality.

Examples: the elementary power series

Exponential. Every derivative of e^x equals e^x , so every derivative at zero equals 1. Therefore

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots. \quad (1.1.5)$$

For fixed $x \neq 0$, the ratio of consecutive absolute terms is

$$\frac{|x|^{k+1}/(k+1)!}{|x|^k/k!} = \frac{|x|}{k+1} \rightarrow 0.$$

The series converges for every x . For real x , it represents e^x because

$$|R_N(x)| \leq e^{|x|} \frac{|x|^{N+1}}{(N+1)!} \rightarrow 0.$$

For example, at $x = 0.1$ the first three terms give $e^{0.1} \approx 1 + 0.1 + 0.005 = 1.105$.

Examples: the elementary power series (cont.)

Cosine and sine. List the derivative values at zero and divide by the corresponding factorial:

k	0	1	2	3	4	5
$(\cos)^{(k)}(0)$	1	0	-1	0	1	0
$(\sin)^{(k)}(0)$	0	1	0	-1	0	1

Only even powers survive for cosine; only odd powers survive for sine:

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad (1.1.6)$$

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots. \quad (1.1.7)$$

For real x , their Taylor errors obey $|R_N(x)| \leq |x|^{N+1}/(N+1)! \rightarrow 0$ because all derivatives have magnitude at most 1. All three series also converge for every complex argument. We use them to extend e^x , $\cos x$, and $\sin x$ to complex numbers.

Imaginary unit, Cartesian and polar form

Define the imaginary unit i by $i^2 = -1$; the other square root of -1 is $-i$. A complex number has Cartesian form

$$z = x + iy, \quad x = \operatorname{Re} z, \quad y = \operatorname{Im} z,$$

where $x, y \in \mathbb{R}$; $\operatorname{Re} z$ and $\operatorname{Im} z$ denote the real and imaginary parts of z . The powers of i repeat with period four:

$$i^{4q} = 1, \quad i^{4q+1} = i, \quad i^{4q+2} = -1, \quad i^{4q+3} = -i.$$

Substitute $i\theta$ into the absolutely convergent exponential series and separate even and odd powers:

$$\begin{aligned} e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= \sum_{k=0}^{\infty} \frac{i^{2k} \theta^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{i^{2k+1} \theta^{2k+1}}{(2k+1)!} \end{aligned}$$

and since $i^{2k} = (i^2)^k = (-1)^k$ and $i^{2k+1} = i i^{2k} = i(-1)^k$,

$$= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!}.$$

Imaginary unit, Cartesian and polar form (cont.)

Comparison with (1.1.6)–(1.1.7) gives Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (1.2.1)$$

The exponential addition law also holds for complex arguments:

$$e^{w_1+w_2} = e^{w_1} e^{w_2}.$$

It follows by multiplying the two absolutely convergent series and applying the binomial theorem. In particular,

$$e^{u+iv} = e^u (\cos v + i \sin v), \quad u, v \in \mathbb{R}. \quad (1.2.2)$$

The factor e^u determines the modulus, and v determines the angle.

For $z \neq 0$, set

$$r = |z| = \sqrt{x^2 + y^2} > 0.$$

Choose any angle θ satisfying $x = r \cos \theta$ and $y = r \sin \theta$. Then

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}. \quad (1.2.3)$$

The angle is not unique: if θ is one argument, then every argument is $\theta + 2\pi k$, $k \in \mathbb{Z}$. The principal argument is commonly chosen as $\text{Arg } z \in (-\pi, \pi]$ (other endpoint conventions are also used). The number $z = 0$ has modulus 0 but no defined argument.

Imaginary unit, Cartesian and polar form (cont.)

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then the phases add by the addition law above, so

$$z_1 z_2 = r_1 r_2 e^{i\theta_1} e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)},$$

so moduli multiply and arguments add modulo 2π .

The complex conjugate is

$$\bar{z} = x - iy = re^{-i\theta}.$$

Consequently

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = r^2 = |z|^2,$$

and adding or subtracting z and $\bar{z}$ isolates the Cartesian components:

$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \quad \operatorname{Im} z = \frac{z - \bar{z}}{2i}. \quad (1.2.4)$$

Replacing θ by $-\theta$ in Euler's formula gives $e^{-i\theta} = \cos \theta - i \sin \theta$. Hence

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta,$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta,$$

and therefore

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (1.2.5)$$

Multivalued functions and branch cuts

An inverse relation need not define a single-valued function. For example, $y = x^2$ gives $x = \pm\sqrt{y}$ for $y > 0$. A *branch* selects one value continuously on a chosen domain; the usual real principal square root selects $+\sqrt{y}$ for $y \geq 0$.

The complex logarithm has the same issue: several inputs have the same exponential. **The complex logarithm.** Let $z \neq 0$ and write $z = re^{i\theta}$ with $r > 0$. To solve $e^w = z$, write $w = u + iv$ with $u, v \in \mathbb{R}$. The splitting (1.2.2) gives

$$e^w = e^{u+iv} = e^u(\cos v + i \sin v).$$

Equality with $r(\cos \theta + i \sin \theta)$ requires

$$e^u = r \implies u = \log r,$$

and

$$\cos v = \cos \theta, \quad \sin v = \sin \theta \implies v = \theta + 2\pi k, \quad k \in \mathbb{Z}.$$

Thus the complete multivalued logarithm is

$$\log z = \log |z| + i(\theta + 2\pi k), \quad k \in \mathbb{Z}. \quad (1.2.6)$$

A principal branch is obtained by choosing one interval of length 2π , commonly

$$-\pi < \operatorname{Arg} z < \pi, \quad \operatorname{Log} z = \log |z| + i \operatorname{Arg} z.$$

This analytic branch is defined on $\mathbb{C} \setminus (-\infty, 0]$. The values $\pm\pi$ are limits from the two sides of the cut, not values on this domain.

Multivalued functions and branch cuts (cont.)

A continuous argument increases by 2π after one counterclockwise turn around 0. Thus the logarithm does not return to its starting value: 0 is a *branch point*. Removing a *branch cut* prevents such a loop from staying in our domain. For the principal logarithm, the cut is the nonpositive real axis. Its two boundary values are

$$\operatorname{Log}(-r + i0) = \log r + i\pi, \quad \operatorname{Log}(-r - i0) = \log r - i\pi,$$

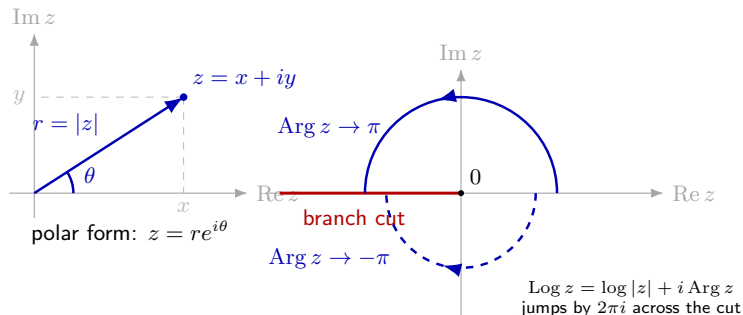
so crossing the cut changes the logarithm by $2\pi i$.

The same mechanism applies to roots. For example,

$$z^{1/2} = \sqrt{r} e^{i(\theta+2\pi k)/2} = (-1)^k \sqrt{r} e^{i\theta/2},$$

so there are two values. A branch cut permits one of them to be selected continuously.

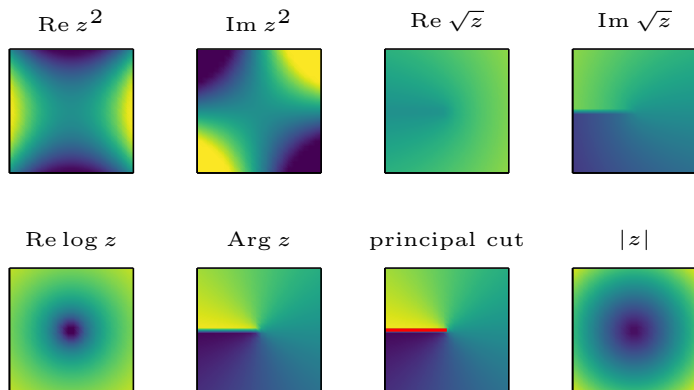
Pictures: branch cuts and component plots



Across the removed ray, the two boundary values of the chosen argument differ by 2π . The branch is defined on the rest of the cut plane.

Pictures: branch cuts and component plots (cont.)

A complex map $z = x + iy \mapsto f = u + iv$ has two real inputs and two real outputs, so we use cut planes and component plots rather than a single graph $y = f(x)$.



The complex derivative and Cauchy–Riemann

The complex derivative at z_0 is

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}, \quad (1.3.1)$$

The increment h may approach zero from any direction in the complex plane. The same limit must result in every case. A function is *holomorphic* (analytic) on an open set if this derivative exists at every point there; it is *entire* if that set is all of $\mathbb{C}$.

Write $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$. Here $u_x = \partial u / \partial x$ means differentiate in x while holding y fixed; u_y , v_x , and v_y are defined in the same way.

Along a real increment $h = \Delta x$, the difference quotient is

$$\begin{aligned} \frac{f(z + h) - f(z)}{h} &= \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \\ &\xrightarrow{\Delta x \rightarrow 0} u_x + iv_x, \end{aligned}$$

so $f'(z) = u_x + iv_x$. Along an imaginary increment $h = i\Delta y$ the bookkeeping is the same, with the extra factor $1/i$ pulled out of the denominator:

$$\frac{f(z + i\Delta y) - f(z)}{i\Delta y} = \frac{1}{i} \left[\frac{u(x, y + \Delta y) - u(x, y)}{\Delta y} + i \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} \right].$$

The complex derivative and Cauchy–Riemann (cont.)

If the complex derivative exists, both directional calculations must give that derivative. Passing to the limit and using $1/i = -i$,

$$f'(z) = \frac{u_y + iv_y}{i} = -i(u_y + iv_y) = v_y - iu_y.$$

Because f is complex differentiable, every direction of approach must return the *same* value $f'(z)$; equating $u_x + iv_x = v_y - iu_y$ and matching real and imaginary parts gives the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x. \quad (1.3.2)$$

They are necessary for complex differentiability.

Working test. If u and v have continuous first partial derivatives near a point, the Cauchy–Riemann equations are also sufficient there, and

$$f'(z) = u_x + iv_x = v_y - iu_y.$$

Thus our procedure is: find u, v , compute four partial derivatives, and compare.

Example: z^2 . Since

$$z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy),$$

we have

$$u_x = 2x = v_y, \quad u_y = -2y = -v_x.$$

Thus z^2 is entire and $f'(z) = 2x + i(2y) = 2z$.

The complex derivative and Cauchy–Riemann (cont.)

Counterexample: $\bar{z} = x - iy$. Then $u_x = 1 \neq -1 = v_y$, so CR fails everywhere. Equivalently $\bar{h}/h$ is 1 for real h and -1 for imaginary h : no direction-independent derivative.

Keep in mind that an analytic function is infinitely differentiable and equals its Taylor series on every disk where it is analytic; we prove this once contour integrals are available.

Contour integrals and a key circle integral

A contour integral is an ordinary integral along a chosen path. Parameterize the path by $z = \gamma(t)$, $a \leq t \leq b$, with γ piecewise continuously differentiable. Since $dz = \gamma'(t) dt$,

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt. \quad (1.4.1)$$

Reversing the path reverses the sign:

$$\int_{-\gamma} f(z) dz = - \int_{\gamma} f(z) dz.$$

A closed contour may be traversed in either direction. A positively oriented boundary keeps the region on the left: the outer boundary is counterclockwise, while a boundary around a hole is clockwise.

Integer powers on a circle. Let $n \in \mathbb{Z}$ and parameterize the positively oriented circle $|z| = R$ by

$$z(\theta) = Re^{i\theta}, \quad dz = iRe^{i\theta} d\theta, \quad 0 \leq \theta \leq 2\pi.$$

Contour integrals and a key circle integral (cont.)

Then

$$\begin{aligned}\oint_{|z|=R} z^n dz &= \int_0^{2\pi} R^n e^{in\theta} (iRe^{i\theta}) d\theta \\ &= iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta.\end{aligned}$$

If $n = -1$, the exponential is 1, so

$$\oint_{|z|=R} \frac{dz}{z} = i \int_0^{2\pi} d\theta = 2\pi i.$$

If $n \neq -1$, then

$$\begin{aligned}iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta &= iR^{n+1} \left[\frac{e^{i(n+1)\theta}}{i(n+1)} \right]_0^{2\pi} \\ &= \frac{R^{n+1}}{n+1} (e^{2\pi i(n+1)} - 1) = 0,\end{aligned}$$

because $n+1$ is an integer. Let $\delta_{n,-1}$ denote the Kronecker delta: it is 1 when $n = -1$ and 0 otherwise. Translating the circle gives

$$\oint_{|z-z_0|=R} (z-z_0)^n dz = 2\pi i \delta_{n,-1}. \quad (1.4.2)$$

Cauchy's integral theorem and formula

A *domain* is an open, connected set. A domain is *simply connected* if every closed curve in it can be continuously contracted to a point without leaving the domain; for the planar regions used here, this is the precise version of “the region has no holes.”

Cauchy's integral theorem

Let D be a simply connected domain, let f be holomorphic on D , and let γ be a closed piecewise smooth contour in D . Then

$$\oint_{\gamma} f(z) dz = 0.$$

Why zero? A direct consequence of Green's theorem and the Cauchy–Riemann equations. With $f = u + iv$ and $dz = dx + i dy$,

$$\begin{aligned}\oint_{\gamma} f dz &= \oint_{\gamma} (u dx - v dy) + i \oint_{\gamma} (v dx + u dy) \\ &= \iint_E (-v_x - u_y) dA + i \iint_E (u_x - v_y) dA \\ &= \iint_E (u_y - u_y) dA + i \iint_E (v_y - v_y) dA = 0.\end{aligned}$$

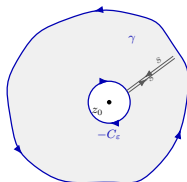
Cauchy's integral theorem and formula (cont.)

Cauchy's integral formula

Let γ be a positively oriented simple closed contour, let f be holomorphic on an open set containing γ and its interior, and let z_0 lie inside γ . Then

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz. \quad (1.4.3)$$

To understand the formula, deform γ to a small counterclockwise circle C_ε about z_0 . The integrand is holomorphic between the two curves, so their integrals agree. The two sides of the connecting slit cancel.



Slit construction: the combined contour $\gamma + s - C_\varepsilon - s$ encloses the shaded region, where the integrand is holomorphic; the inner circle is traversed clockwise and the slit sides s and $-s$ cancel.

Cauchy's integral theorem and formula (cont.)

On the small circle $z = z_0 + \varepsilon e^{i\theta}$,

$$\oint_{C_\varepsilon} \frac{f(z)}{z - z_0} dz = i \int_0^{2\pi} f(z_0 + \varepsilon e^{i\theta}) d\theta.$$

As $\varepsilon \rightarrow 0$, continuity makes the integrand approach $f(z_0)$, so the last integral approaches $2\pi i f(z_0)$. This gives (1.4.3).

Differentiate (1.4.3) in z_0 (contour fixed, away from z_0). Since $\partial_{z_0}^n (z - z_0)^{-1} = n! / (z - z_0)^{n+1}$,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz. \quad (1.4.4)$$

In particular every holomorphic function is infinitely differentiable, and

$$\oint_\gamma \frac{f(z)}{(z - z_0)^m} dz = \frac{2\pi i}{(m-1)!} f^{(m-1)}(z_0).$$

Taylor series from contour integration

Taylor series theorem

If f is holomorphic on the disk $|z - z_0| < R$, then for every $|z - z_0| < R$,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

Choose a radius ρ satisfying

$$|z - z_0| < \rho < R$$

and integrate on $C_\rho : |\xi - z_0| = \rho$. Cauchy's formula gives

$$f(z) = \frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{\xi - z} d\xi.$$

The idea is to expand only the kernel, using the geometric series:

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \frac{1}{1 - (z - z_0)/(\xi - z_0)} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}.$$

Taylor series from contour integration (cont.)

This works because $|(z - z_0)/(\xi - z_0)| = |z - z_0|/\rho < 1$ all along the circle. Substituting and integrating the convergent series term by term gives

$$f(z) = \sum_{n=0}^{\infty} (z - z_0)^n \underbrace{\frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi}_{f^{(n)}(z_0)/n! \text{ by (1.4.4)}}.$$

The coefficients are exactly those found by matching derivatives at the start of the chapter. Here Cauchy's formula also tells us where the series equals f .

The key consequence is simple: inside a disk where f is holomorphic, its Taylor series represents f . For an expansion about z_0 , the nearest nonremovable singularity limits the radius of convergence. Boundary points require separate checking.

The converse also holds: a convergent power series may be differentiated term by term inside its disk, so it is holomorphic there. Being analytic at z_0 and having a Taylor series representation about z_0 are the same property.

Generalized binomial series. Take $f(z) = (1 + z)^m$. For noninteger m , choose $(1 + z)^m = e^{m \operatorname{Log}(1+z)}$, which is single valued for $|z| < 1$ and equals 1 at $z = 0$. Successive derivatives are

$$f'(z) = m(1 + z)^{m-1}, \quad f''(z) = m(m-1)(1 + z)^{m-2},$$

Taylor series from contour integration (cont.)

so the pattern is

$$f^{(n)}(z) = m(m-1)\cdots(m-n+1)(1+z)^{m-n}.$$

At $z = 0$,

$$\frac{f^{(n)}(0)}{n!} = \frac{m(m-1)\cdots(m-n+1)}{n!} = \binom{m}{n}.$$

Hence

$$(1+z)^m = \sum_{n=0}^{\infty} \binom{m}{n} z^n = 1 + mz + \frac{m(m-1)}{2!} z^2 + \frac{m(m-1)(m-2)}{3!} z^3 + \cdots. \quad (1.5.1)$$

If $m \in \{0, 1, 2, \dots\}$, the factor $m - n + 1$ becomes zero and the series terminates, so the result is a polynomial valid for every z . Otherwise the nearest obstruction to the expansion about 0 is $z = -1$: it is a pole when m is a negative integer and a branch point when m is nonintegral. Therefore the radius of convergence is 1.

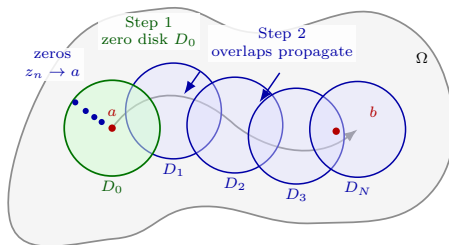
Analytic continuation

A formula may stop working before the function itself does. For example,

$$1 + z + z^2 + \cdots = \frac{1}{1 - z} \quad (|z| < 1).$$

The series fails outside that disk, but $1/(1 - z)$ remains holomorphic everywhere except $z = 1$. It *continues* the same function beyond the original series.

More generally, overlapping Taylor disks let us carry a function to a larger domain:



The formulas must agree on each overlap; they are not unrelated approximations.

Analytic continuation (cont.)

Identity theorem: the uniqueness principle

If two holomorphic functions on a connected domain agree at infinitely many points approaching a point inside that domain, they agree throughout the domain. In particular, agreement on an open overlap is enough.

Agreement on an open overlap therefore fixes the continuation on a connected common domain. We will use this to extend Gamma-function identities.

There is one important limitation: following different paths around a branch point can lead to different branches. For $\log z$, one counterclockwise turn about 0 adds $2\pi i$. A fixed branch cut removes this ambiguity from our calculations.

Laurent series

A Taylor series uses nonnegative powers. Near a singularity we also need $1/(z - z_0)$, $1/(z - z_0)^2$, $\dots$. Their combination is a *Laurent series*.

Suppose f is holomorphic in the annulus

$$r < |z - z_0| < R.$$

An inner hole allows negative powers; the next singularity limits the outer radius. The Laurent theorem gives the representation below. We will obtain its coefficients and use geometric series to construct examples, rather than prove the full theorem. Thus f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n = \underbrace{\sum_{n=0}^{\infty} a_n(z - z_0)^n}_{\text{regular part}} + \underbrace{\sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}}_{\text{principal part}}, \quad (1.5.2)$$

Laurent series (cont.)

convergent throughout the annulus. To extract a coefficient, multiply by $(z - z_0)^{-k-1}$ and integrate on a centered circle $\gamma : |z - z_0| = \rho$, where $r < \rho < R$, oriented counterclockwise:

$$\begin{aligned}\frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{k+1}} dz &= \frac{1}{2\pi i} \sum_{n=-\infty}^{\infty} a_n \oint_{\gamma} (z - z_0)^{n-k-1} dz \\ &= \frac{1}{2\pi i} a_k (2\pi i) = a_k.\end{aligned}$$

Therefore, for every integer k ,

$$a_k = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{k+1}} dz. \quad (1.5.3)$$

On the chosen circle inside the annulus, term-by-term integration is valid.

If the principal part is empty, the series extends holomorphically to the full disk and is a Taylor series. A punctured disk $0 < |z - z_0| < R$ does not guarantee this: $1/(z - z_0)$ is the simplest counterexample.

Laurent series (cont.)

Coefficient extraction. On $0 < |z| < 1$,

$$f(z) = \frac{1}{z(z-1)} = \frac{1}{z-1} - \frac{1}{z} = -\frac{1}{z} - \sum_{n=0}^{\infty} z^n,$$

so $a_{-1} = -1$ and $a_n = -1$ for $n \geq 0$. Only a_{-1} survives when f itself is integrated around a small circle about the singularity. It is called the *residue*, written $\text{Res}(f, 0) = -1$; we develop it in §1.7. On $|z| > 1$ the geometric expansion changes. Always specify the annulus before reading off Laurent coefficients.

Removable example: sinc. Dividing the sine series by z gives

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n+1)!},$$

entire after setting $\text{sinc } 0 = 1$.

Classifying singularities

A point z_0 is a *singular point* of f when f fails to be holomorphic there. It is an *isolated singularity* if there exists $R > 0$ such that f is holomorphic on the punctured disk

$$0 < |z - z_0| < R.$$

Only isolated singularities admit an ordinary Laurent classification. Write

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

Then:

- The singularity is **removable** if $a_n = 0$ for every $n < 0$. Defining $f(z_0) = a_0$ makes f holomorphic at z_0 . Example: $\sin z/z$ at 0.
- It is a **pole of order m** , where $m \geq 1$ is an integer, if $a_{-m} \neq 0$ and $a_n = 0$ for every $n < -m$. Equivalently,

$$f(z) = \frac{h(z)}{(z - z_0)^m}, \quad h \text{ holomorphic, } h(z_0) \neq 0.$$

Multiplying by $(z - z_0)^m$ removes exactly the highest negative power.

Classifying singularities (cont.)

- It is **essential** if infinitely many negative-power coefficients are nonzero. For example,

$$\sin\left(\frac{1}{z}\right) = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \cdots$$

has an essential singularity at 0.

A function holomorphic except for isolated poles is *meromorphic*. For example, $1/\sin z$ is meromorphic, with simple poles at $z = n\pi$, $n \in \mathbb{Z}$.

To check this example, $\sin z = 0$ is equivalent to $e^{2iz} = 1$. Writing $z = x + iy$, its modulus gives $e^{-2y} = 1$, so $y = 0$; its phase then gives $x = n\pi$. Near such a point, Taylor expansion gives

$$\sin z = \cos(n\pi)(z - n\pi) + \cdots = (-1)^n(z - n\pi) + \cdots.$$

Thus $1/\sin z$ begins with $(-1)^n/(z - n\pi)$ and has a simple pole there.

A branch point is not an isolated singularity. For example, analytic continuation of $\log z$ once around 0 changes its value by $2\pi i$, so no single-valued branch is holomorphic on a complete punctured disk around 0. The removable/pole/essential trichotomy therefore does not apply to the branch point itself.

At infinity, inspect $F(w) = f(1/w)$ at $w = 0$. A polynomial of degree $m \geq 1$ has a pole of order m there; a constant is removable; e^z is essential.

The residue theorem

For an isolated singularity z_0 , the residue is the Laurent coefficient

$$\operatorname{Res}(f, z_0) = a_{-1}.$$

Choose a small counterclockwise circle C about z_0 enclosing no other singularity. Use the Laurent expansion valid near z_0 , not one on an annulus containing other singularities. By (1.4.2), integrating any integer power around C gives zero except for $(z - z_0)^{-1}$, whose integral is $2\pi i$. Only the $n = -1$ term survives:

$$\begin{aligned}\oint_C f(z) dz &= \sum_{n=-\infty}^{\infty} a_n \oint_C (z - z_0)^n dz \\ &= a_{-1}(2\pi i) \\ &= 2\pi i \operatorname{Res}(f, z_0).\end{aligned}$$

Thus

$$\oint_C f(z) dz = 2\pi i \operatorname{Res}(f, z_0). \quad (1.7.1)$$

The residue theorem (cont.)

Residue theorem

Let f be holomorphic inside and on a positively oriented simple closed contour γ , except for finitely many isolated singularities $z_1, \dots, z_N$ in the interior. Then

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^N \operatorname{Res}(f, z_k). \quad (1.7.2)$$

Geometric reason. Deform the outer contour into small counterclockwise circles around the singularities. Each circle contributes its own $2\pi i$ times residue. The simple outer contour encloses each of these singularities once.

The residue theorem (cont.)

Residue at a pole of order m . Write

$$h(z) = (z - z_0)^m f(z),$$

where h is holomorphic and $h(z_0) \neq 0$. Taylor expansion gives

$$h(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^j.$$

Dividing by $(z - z_0)^m$,

$$f(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^{j-m}.$$

The exponent equals -1 exactly when $j = m - 1$. Therefore

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \bigg|_{z=z_0}. \quad (1.7.3)$$

For $m = 1$ this reduces to

$$\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z),$$

and for $m = 2$ it becomes the first derivative of $(z - z_0)^2 f(z)$ at z_0 .

The residue theorem (cont.)

A quotient at a simple zero. If $q(z_0) = 0$ and $q'(z_0) \neq 0$, then

$$q(z) = q'(z_0)(z - z_0) + \cdots.$$

For p, q holomorphic near z_0 , cancel $z - z_0$ before taking the limit:

$$\operatorname{Res}\left(\frac{p}{q}, z_0\right) = \lim_{z \rightarrow z_0} \frac{p(z)}{q(z)/(z - z_0)} = \frac{p(z_0)}{q'(z_0)}. \quad (1.7.4)$$

When $p(z_0) = 0$, the singularity is removable and the formula gives zero.

For example,

$$\operatorname{Res}\left(\frac{1}{z^2 + 1}, i\right) = \frac{1}{2i},$$

while the order-2 formula at the double pole of $1/(z^2 + 1)^2$ gives

$$\operatorname{Res}\left(\frac{1}{(z^2 + 1)^2}, i\right) = \left. \frac{d}{dz} \frac{1}{(z + i)^2} \right|_{z=i} = -\frac{2}{(2i)^3} = \frac{1}{4i}.$$

The residue theorem (cont.)

Quick checks on $|z| = 1$.

$$\frac{\sin z}{z^2} = \frac{1}{z} - \frac{z}{3!} + \cdots \implies \operatorname{Res}\left(\frac{\sin z}{z^2}, 0\right) = 1,$$

so $\oint_{|z|=1} (\sin z)/z^2 dz = 2\pi i$. By contrast $z^2/\sin z$ extends holomorphically through 0 (value 0) and has no other zeros of $\sin z$ inside the unit disk, so

$$\oint_{|z|=1} \frac{z^2}{\sin z} dz = 0.$$

Real integrals by a semicircular contour

Let Γ_R consist of the real segment from $-R$ to R and the upper semicircle $C_R : z = Re^{i\theta}$, $0 \leq \theta \leq \pi$. For an integrand with no pole on the real segment,

$$\oint_{\Gamma_R} f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz. \quad (1.8.1)$$

The magnitude of an integral is at most the maximum integrand magnitude M times the path length L ; this is the ML estimate. If $M_R = \max_{z \in C_R} |f(z)|$, the arc length is πR , so

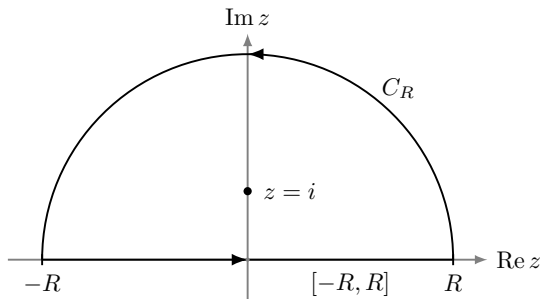
$$\left| \int_{C_R} f(z) dz \right| \leq \pi R M_R.$$

Thus the arc vanishes whenever $RM_R \rightarrow 0$. If the improper real integral converges, the residue theorem then gives

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}(f, z_k).$$

When an exponential e^{iaz} is present, its decay in one half-plane often supplies the needed estimate.

Real integrals by a semicircular contour (cont.)



Real integrals by a semicircular contour (cont.)

Example: $\int_{-\infty}^{\infty} \cos x / (1 + x^2) dx$. Consider

$$J = \int_{-\infty}^{\infty} \frac{e^{ix}}{1 + x^2} dx.$$

For $z = x + iy$ in the upper half-plane,

$$|e^{iz}| = |e^{ix-y}| = e^{-y} \leq 1.$$

On $|z| = R > 1$,

$$|1 + z^2| \geq ||z|^2 - 1| = R^2 - 1,$$

so

$$\left| \int_{C_R} \frac{e^{iz}}{1 + z^2} dz \right| \leq \frac{\pi R}{R^2 - 1} \rightarrow 0.$$

Only the pole $z = i$ lies in the upper half-plane. Since $1 + z^2 = (z - i)(z + i)$,

$$\begin{aligned} \operatorname{Res}\left(\frac{e^{iz}}{1 + z^2}, i\right) &= \frac{e^{iz}}{z + i} \Big|_{z=i} \\ &= \frac{e^{-1}}{2i}. \end{aligned}$$

Real integrals by a semicircular contour (cont.)

Hence

$$J = 2\pi i \frac{e^{-1}}{2i} = \frac{\pi}{e}.$$

Taking real parts gives

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx = \frac{\pi}{e}. \quad (1.8.2)$$

The imaginary part vanishes consistently because $\sin x/(1+x^2)$ is odd.

Example: $\int_{-\infty}^{\infty} dx/(1+x^4)$. Close in the upper half-plane. The upper arc vanishes by ML: $|\int_{C_R}| \leq \pi R/(R^4-1) \rightarrow 0$ as $R \rightarrow \infty$. The poles are the fourth roots of -1 :

$$z_k = e^{i(\pi+2k\pi)/4}, \quad k = 0, 1, 2, 3.$$

Those in the upper half-plane are the simple poles

$$z_0 = e^{i\pi/4}, \quad z_1 = e^{i3\pi/4}.$$

Since $1+z^4 = \prod_{k=0}^3 (z-z_k)$ and $(1+z^4)' = 4z^3$, each residue is

$$\text{Res}\left(\frac{1}{1+z^4}, z_j\right) = \frac{1}{4z_j^3}.$$

Real integrals by a semicircular contour (cont.)

Since $z_j^4 = -1$, this residue is $-z_j/4$. Adding the two upper-half-plane residues,

$$\sum_{j=0}^1 \operatorname{Res}\left(\frac{1}{1+z^4}, z_j\right) = -\frac{e^{i\pi/4} + e^{3i\pi/4}}{4} = -\frac{i}{2\sqrt{2}}.$$

The residue theorem therefore gives

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = 2\pi i \left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}. \quad (1.8.3)$$

Keyhole contour: $\int_0^\infty x^a / (1+x)^2 dx$

Let

$$I(a) = \int_0^\infty \frac{x^a}{(1+x)^2} dx, \quad -1 < a < 1.$$

The endpoint conditions follow directly: near 0 the integrand behaves as x^a , which is integrable when $a > -1$; near infinity it behaves as x^{a-2} , which is integrable when $a < 1$.

For $a \neq 0$, choose the branch of the power whose cut runs along the *positive* real axis,

$$z^a = e^{a\ell(z)}, \quad \ell(z) = \log |z| + i \arg z, \quad 0 < \arg z < 2\pi,$$

rather than the principal logarithm used earlier. The cut is chosen along the real integration interval so that the keyhole follows its two sides. Use a counterclockwise keyhole with outer radius R and inner radius ε . The hole keeps the branch point 0 outside the enclosed region. A branch point has no ordinary residue: the power is not single-valued on a full punctured disk.

On the upper side, $z = x$ and the contribution is

$$I_{\varepsilon,R}(a) = \int_\varepsilon^R \frac{x^a}{(1+x)^2} dx.$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

On the lower side, $z = xe^{2\pi i}$ and $z^a = x^a e^{2\pi ia}$. The orientation is from R back to ε , so the contribution is

$$\int_R^\varepsilon \frac{x^a e^{2\pi ia}}{(1+x)^2} dx = -e^{2\pi ia} I_{\varepsilon,R}(a).$$

Thus the two straight segments sum to

$$(1 - e^{2\pi ia}) I_{\varepsilon,R}(a).$$

We write $F = O(G)$ when $|F|$ is bounded by a constant times $|G|$ in the stated limit. On the outer circle, since a is real and $z^a = e^{a(\log R + i \arg z)}$, one has $|z^a| = R^a$. Hence

$$\left| \frac{z^a}{(1+z)^2} \right| \leq \frac{R^a}{(R-1)^2} = O(R^{a-2}),$$

so the arc integral is $O(R^{a-1}) \rightarrow 0$ because $a < 1$. On the inner circle,

$$\left| \frac{z^a}{(1+z)^2} \right| = O(\varepsilon^a),$$

so the arc integral is $O(\varepsilon^{a+1}) \rightarrow 0$ because $a > -1$.

The only enclosed singularity is the double pole at $z = -1$. Since

$$(z+1)^2 \frac{z^a}{(1+z)^2} = z^a,$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

formula (1.7.3) gives

$$\operatorname{Res}\left(\frac{z^a}{(1+z)^2}, -1\right) = \left.\frac{d}{dz}z^a\right|_{z=-1}.$$

The chain rule gives $(z^a)' = az^{a-1}$ on this branch. Since $\ell(-1) = i\pi$,

$$\operatorname{Res}\left(\frac{z^a}{(1+z)^2}, -1\right) = ae^{i\pi(a-1)} = -ae^{i\pi a}.$$

Hence the residue theorem gives, after $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$,

$$(1 - e^{2\pi ia})I(a) = -2\pi iae^{i\pi a}.$$

Factor the denominator:

$$\begin{aligned}1 - e^{2\pi ia} &= e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) \\&= -2ie^{i\pi a}\sin(\pi a).\end{aligned}$$

Therefore

$$\begin{aligned}I(a) &= \frac{-2\pi iae^{i\pi a}}{-2ie^{i\pi a}\sin(\pi a)} \\&= \frac{\pi a}{\sin(\pi a)}.\end{aligned}$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ (cont.)

Thus

$$\int_0^\infty \frac{x^a}{(1+x)^2} dx = \frac{\pi a}{\sin(\pi a)}, \quad -1 < a < 1. \quad (1.8.4)$$

At $a = 0$, the right-hand side has limit 1 because $\sin(\pi a) \sim \pi a$, where $u(a) \sim v(a)$ means $u(a)/v(a) \rightarrow 1$ in the stated limit, and directly $\int_0^\infty (1+x)^{-2} dx = 1$.

The same keyhole with the simpler integrand $z^{-a}/(1+z)$ (simple pole at -1) yields the companion evaluation, valid for $0 < a < 1$,

$$\int_0^\infty \frac{x^{-a}}{1+x} dx = \frac{\pi}{\sin(\pi a)}. \quad (1.8.5)$$

Here the lower-side phase is $e^{-2\pi ia}$, and the residue at -1 is $e^{-i\pi a}$. The outer and inner arcs are respectively $O(R^{-a})$ and $O(\varepsilon^{1-a})$, so both vanish. Therefore

$$(1 - e^{-2\pi ia}) \int_0^\infty \frac{x^{-a}}{1+x} dx = 2\pi i e^{-i\pi a}.$$

Using $1 - e^{-2\pi ia} = 2ie^{-i\pi a} \sin(\pi a)$ gives (1.8.5). This reflection factor reappears in the Gamma and Beta functions.

Jordan's lemma and $\int \sin x/x \, dx$

A small clockwise semicircle around a real-axis pole. Near a simple pole z_0 , write

$$f(z) = \frac{r_{-1}}{z - z_0} + g(z), \quad r_{-1} = \text{Res}(f, z_0),$$

with g bounded. Parameterize the upper indentation by $z = z_0 + \varepsilon e^{i\theta}$, traversed from $\theta = \pi$ to 0. Then

$$\frac{r_{-1}}{z - z_0} dz = ir_{-1} d\theta, \quad ir_{-1} \int_{\pi}^0 d\theta = -i\pi r_{-1}.$$

The bounded part contributes at most its bound times $\pi\varepsilon$, so

$$\int_{C_\varepsilon} f(z) dz \longrightarrow -i\pi \text{Res}(f, z_0). \quad (1.8.6)$$

The minus sign comes from the clockwise orientation.

Jordan-type semicircle estimate

Let C_R be the upper semicircle and let $M_R = \max_{z \in C_R} |f(z)|$. If $M_R \rightarrow 0$ as $R \rightarrow \infty$, then for every $a > 0$,

$$\int_{C_R} f(z) e^{iaz} dz \longrightarrow 0.$$

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

We use this estimate without its full proof. On $z = Re^{i\theta}$, $|e^{iaz}| = e^{-aR \sin \theta}$, and the angular integral satisfies

$$\left| \int_{C_R} f(z) e^{iaz} \, dz \right| \leq \frac{\pi M_R}{a}.$$

The estimate includes the small regions near $\theta = 0, \pi$, where the exponential alone does not decay. Continuity on the arc suffices for the estimate; analyticity is needed separately when applying the residue theorem.

Example: the Dirichlet integral. Integrate e^{iz}/z over the upper semicircle, indenting clockwise above $z = 0$. On the large arc, take $f(z) = 1/z$, so $M_R = 1/R \rightarrow 0$ and the Jordan estimate applies. The small semicircle over $z = 0$ is traversed clockwise through angle π , so by the indentation rule (1.8.6) it contributes $-i\pi \operatorname{Res}$:

$$\begin{aligned} -i\pi \operatorname{Res}\left(\frac{e^{iz}}{z}, 0\right) &= -i\pi \lim_{z \rightarrow 0} e^{iz} \\ &= -i\pi. \end{aligned}$$

The indentation excludes the only pole, so e^{iz}/z is holomorphic on and inside the indented contour and Cauchy's theorem makes the total integral zero. Written out,

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

that contour has four pieces — the segment $[-R, -\varepsilon]$, the small semicircle C_ε traversed clockwise from $-\varepsilon$ to $+\varepsilon$, the segment $[\varepsilon, R]$, and the large semicircle C_R :

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx + \int_{C_\varepsilon} \frac{e^{iz}}{z} dz + \int_{\varepsilon}^R \frac{e^{ix}}{x} dx + \int_{C_R} \frac{e^{iz}}{z} dz = 0.$$

The two real segments must be combined symmetrically because $\cos x/x$ behaves as $1/x$ near zero. This is the *Cauchy principal value*:

$$\text{PV} \int_{-\infty}^{\infty} g(x) dx = \lim_{\substack{\varepsilon \rightarrow 0^+ \\ R \rightarrow \infty}} \left(\int_{-R}^{-\varepsilon} g(x) dx + \int_{\varepsilon}^R g(x) dx \right).$$

The large arc tends to 0 and the small arc to $-i\pi$. Letting $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$ in the four-piece identity therefore gives

$$0 = \text{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx - i\pi,$$

and hence

$$\text{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx = i\pi. \quad (1.8.7)$$

Jordan's lemma and $\int \sin x/x \, dx$ (cont.)

Taking imaginary parts,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx = \pi. \quad (1.8.8)$$

The integrand $\sin x/x$ is continuous at 0 after assigning the value 1, so its integral needs no principal-value interpretation there; the principal value was required only for the auxiliary exponential integral.

Sampling kernels: writing a sum as an integral

Residue calculus also runs in the other direction: a *sum* can be written as a contour integral by multiplying the summand by a kernel whose poles sit at the sample points.

Integer samples. The kernel $\pi \cot \pi z$ has a simple pole at every integer n with residue

$$\operatorname{Res}_{z=n} \pi \cot \pi z = \frac{\pi \cos \pi n}{\pi \cos \pi n} = 1.$$

For f holomorphic inside a contour C enclosing the integers $n_1, \dots, n_2$ (and no other poles of the integrand),

$$\sum_{m=n_1}^{n_2} f(m) = \frac{1}{2\pi i} \oint_C f(z) \pi \cot \pi z \, dz.$$

The alternating variant uses $\pi \csc \pi z = \pi / \sin \pi z$, whose residue at n is $(-1)^n$, giving $\sum (-1)^m f(m)$.

Half-integer samples. The kernel $\pi / \cos \pi z$ has residue $(-1)^{n+1}$ at $z = n + \frac{1}{2}$ (since $(\cos \pi z)'|_{n+1/2} = -\pi(-1)^n$), so a contour enclosing the half-integers samples $\sum (-1)^{n+1} f(n + \frac{1}{2})$.

The payoff comes after the rewrite: deforming C moves the value of the sum onto the poles and saddle contributions of f itself. Turning a slowly converging modal

Sampling kernels: writing a sum as an integral (cont.)

series into residues at complex poles of the summand is the Watson transformation of scattering theory; this frame is its entire complex analysis.

The steepest-descent approximation

Step 1: set up the large-parameter integral. Many asymptotic formulas for special functions come from integrals of the form

$$I(s) = \int_{\Gamma} g(z) e^{sf(z)} dz, \quad s \rightarrow +\infty, \quad (1.9.1)$$

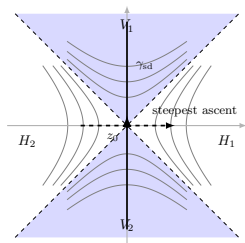
where f and g are holomorphic near a contour Γ that may be deformed without crossing singularities of the integrand. Write $f = u + iv$ with real u, v . As s grows, e^{su} favors points where u is largest on the deformed contour. We study the case where a *saddle* supplies the leading contribution; endpoints must not dominate.

Step 2: locate a nondegenerate saddle. We seek a point z_0 where

$$f'(z_0) = 0, \quad f''(z_0) \neq 0.$$

At such a point the first-order change vanishes. The quadratic term determines which directions make $\operatorname{Re} f$ decrease. In the complex plane these are descent directions through a saddle, rather than a maximum in every direction.

The steepest-descent approximation (cont.)



In the schematic, hill sectors and valley sectors meet at the saddle; a steepest-descent path through the saddle runs into the valleys of $u = \operatorname{Re} f$.

The steepest-descent approximation (cont.)

Step 3: choose the descent direction. Taylor expansion at a nondegenerate saddle is

$$f(z) = f(z_0) + \frac{1}{2}f''(z_0)(z - z_0)^2 + O((z - z_0)^3). \quad (1.9.2)$$

Write

$$f''(z_0) = |f''(z_0)|e^{i\theta}, \quad z - z_0 = re^{i\phi}.$$

Then

$$\frac{1}{2}f''(z_0)(z - z_0)^2 = \frac{1}{2}|f''(z_0)|r^2 e^{i(\theta+2\phi)}.$$

A *steepest-descent* direction makes this quadratic change negative real, so

$$\theta + 2\phi = \pi \pmod{2\pi}, \quad \text{that is} \quad \phi = \frac{\pi - \theta}{2} \pmod{\pi}. \quad (1.9.3)$$

The ambiguity by π is not a defect: it is the two opposite rays leaving z_0 , and Step 4 uses both at once by letting r run over all of $\mathbb{R}$. Along that ray,

$$\frac{1}{2}f''(z_0)(z - z_0)^2 = -\frac{1}{2}|f''(z_0)|r^2.$$

(The orthogonal choice $\theta + 2\phi = 0$ is steepest *ascent*.)

The steepest-descent approximation (cont.)

Step 4: reduce to a Gaussian integral. Assume the contour can be deformed through z_0 along both descent rays, that this saddle dominates the integral, and that $g(z_0) \neq 0$. Parameterize the full descent line by

$$\alpha = e^{i(\pi-\theta)/2} = i e^{-i\theta/2}, \quad z = z_0 + \alpha r, \quad r \in \mathbb{R},$$

Thus $dz = \alpha dr$. Since $\alpha^2 = -e^{-i\theta}$, we have $f''(z_0)\alpha^2 = -|f''(z_0)|$. Replacing f by its quadratic Taylor piece and g by the constant $g(z_0)$ yields

$$\begin{aligned} I(s) &\sim g(z_0) e^{sf(z_0)} \alpha \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}s|f''(z_0)|r^2\right) dr \\ &= g(z_0) e^{sf(z_0)} \alpha \sqrt{\frac{2\pi}{s|f''(z_0)|}}, \end{aligned}$$

where we used $\int_{-\infty}^{\infty} e^{-ar^2} dr = \sqrt{\pi/a}$ with $a = \frac{1}{2}s|f''(z_0)|$. This standard Gaussian integral is evaluated in Chapter 2.

Reverse orientation changes the sign; one descent ray gives half the Gaussian. For several equally dominant saddles, add their contributions. Therefore, for the stated orientation,

$$I(s) \sim g(z_0) e^{sf(z_0)} i e^{-i\theta/2} \sqrt{\frac{2\pi}{s|f''(z_0)|}}. \quad (1.9.4)$$

Real special cases: Laplace method and stationary phase

The same local-Gaussian idea appears in two real-variable forms used later in the course.

Laplace's method (real, decaying exponential). For a smooth real phase φ with a unique interior maximum at x_0 , $\varphi'(x_0) = 0$, $\varphi''(x_0) < 0$, and a smooth amplitude g with $g(x_0) \neq 0$,

$$\int_a^b g(x) e^{s\varphi(x)} dx \sim g(x_0) e^{s\varphi(x_0)} \sqrt{\frac{2\pi}{s|\varphi''(x_0)|}}, \quad s \rightarrow +\infty. \quad (1.9.5)$$

This is (1.9.4) restricted to the real line with $\theta = \pi$ (so $f''(x_0) = -|\varphi''(x_0)|$ is already negative real and $\alpha = 1$). Chapter 2 uses this pattern for Stirling's approximation of $n!$ from the gamma integral.

Stationary phase (real, oscillatory). For a real phase φ with $\varphi'(x_0) = 0$, $\varphi''(x_0) \neq 0$,

$$\int g(x) e^{is\varphi(x)} dx \sim g(x_0) e^{is\varphi(x_0)} e^{i(\pi/4) \operatorname{sgn} \varphi''(x_0)} \sqrt{\frac{2\pi}{s|\varphi''(x_0)|}}, \quad s \rightarrow +\infty, \quad (1.9.6)$$

Here $\operatorname{sgn} q$ is $+1$ for $q > 0$ and -1 for $q < 0$. Assume the smooth amplitude is zero near the endpoints, x_0 is the only stationary point where the amplitude is nonzero,

Real special cases: Laplace method and stationary phase (cont.)

and $g(x_0) \neq 0$. Contributions from several nondegenerate stationary points are added. The Fresnel phase factor arises from the local Gaussian limit. Near x_0 ,

$$\varphi(x) \approx \varphi(x_0) + \frac{1}{2}\varphi''(x_0)(x - x_0)^2.$$

Set $u = \sqrt{s}(x - x_0)$, so $dx = du/\sqrt{s}$. Then

$$\int g(x) e^{is\varphi(x)} dx \sim e^{is\varphi(x_0)} \frac{g(x_0)}{\sqrt{s}} \int_{-\infty}^{\infty} e^{i(\varphi''(x_0)/2)u^2} du.$$

The oscillatory Gaussian has the corresponding formula

$$\int_{-\infty}^{\infty} e^{\pm iau^2} du = \sqrt{\frac{\pi}{a}} e^{\pm i\pi/4}, \quad a > 0.$$

One way to obtain it is to insert a damping factor $e^{-\varepsilon u^2}$, use $\sqrt{\pi/(\varepsilon \mp ia)}$, and let $\varepsilon \rightarrow 0^+$. We use the resulting Fresnel formula without its convergence proof. The positive and negative curvatures give the factors $e^{i\pi/4}$ and $e^{-i\pi/4}$, respectively: together they give (1.9.6). Chapter 5 applies this rule to the unit-circle integral for $J_n(x)$.

Real special cases: Laplace method and stationary phase (cont.)

Two refinements to carry forward. When the dominant point is an *endpoint* rather than an interior saddle or stationary point, repeated integration by parts replaces the Gaussian. The simplest case is *Watson's lemma*: for smooth g with sufficient decay,

$$\int_0^\infty g(t)e^{-st} dt \sim \frac{g(0)}{s} + \frac{g'(0)}{s^2} + \cdots, \quad s \rightarrow +\infty.$$

The endpoint supplies algebraic, not $1/\sqrt{s}$, decay.

Second, (1.9.4) assumes the amplitude g varies slowly near the saddle. If a pole of the integrand approaches or crosses the saddle—the geometry behind a transition region in wave problems—the single-Gaussian answer is no longer uniform, and an error-function-type transition function takes its place. Recognizing which refinement a problem needs is part of using the method.

Worked example: large binomial coefficient

Let m, n be integers with $0 < m < n$. As $n \rightarrow \infty$, keep $t = m/n$ away from 0 and 1. From $(1+z)^n = \sum_{k=0}^n \binom{n}{k} z^k$, Cauchy's coefficient formula gives

$$\binom{n}{m} = \frac{1}{2\pi i} \oint \frac{(1+z)^n}{z^{m+1}} dz = \oint g(z) e^{nf(z)} dz,$$

where

$$g(z) = \frac{1}{2\pi i z}, \quad f(z) = \log(1+z) - t \log z$$

(with branches chosen locally near the positive saddle below).

Saddle. Differentiate:

$$f'(z) = \frac{1}{1+z} - \frac{t}{z}, \quad f''(z) = -\frac{1}{(1+z)^2} + \frac{t}{z^2}.$$

Set $f'(z_0) = 0$:

$$z_0 - t(1+z_0) = 0 \implies z_0 = \frac{t}{1-t} = \frac{m}{n-m} > 0.$$

Worked example: large binomial coefficient (cont.)

Then $1 + z_0 = 1/(1 - t)$, and

$$\begin{aligned}f(z_0) &= \log \frac{1}{1-t} - t \log \frac{t}{1-t} = -t \log t - (1-t) \log(1-t), \\f''(z_0) &= -\frac{1}{(1+z_0)^2} + \frac{t}{z_0^2} = -(1-t)^2 + \frac{(1-t)^2}{t} \\&= (1-t)^2 \left(\frac{1}{t} - 1 \right) = \frac{(1-t)^3}{t} > 0.\end{aligned}$$

Thus $\theta = \arg f''(z_0) = 0$, so the descent direction from (1.9.3) is $\phi = \pi/2$: the local descent line is vertical. Take the contour in Cauchy's coefficient formula to be the circle $|z| = z_0$. On this circle, $|z|$ is fixed and

$$|1 + z_0 e^{i\psi}|^2 = 1 + 2z_0 \cos \psi + z_0^2$$

is largest at $\psi = 0$. The saddle $z = z_0$ therefore dominates. The counterclockwise tangent there points upward ($+i$), matching the descent direction and fixing the sign of the result.

Apply (1.9.4) with $s = n$.

$$\binom{n}{m} \sim \frac{1}{2\pi i z_0} e^{nf(z_0)} i \sqrt{\frac{2\pi}{nf''(z_0)}} = \frac{1}{2\pi z_0} e^{nf(z_0)} \sqrt{\frac{2\pi}{nf''(z_0)}}.$$

Worked example: large binomial coefficient (cont.)

Substitute $1/z_0 = (1-t)/t$ and $1/f''(z_0) = t/(1-t)^3$:

$$\frac{1}{2\pi z_0} \sqrt{\frac{2\pi}{nf''(z_0)}} = \frac{1-t}{2\pi t} \sqrt{\frac{2\pi t}{n(1-t)^3}} = \frac{1}{\sqrt{2\pi nt(1-t)}}.$$

Also $e^{nf(z_0)} = \exp[-n(t \log t + (1-t) \log(1-t))]$, so

$$\binom{n}{m} \sim \frac{\exp[-n(t \log t + (1-t) \log(1-t))]}{\sqrt{2\pi nt(1-t)}}. \quad (1.9.7)$$

Chapter 2's Stirling formula gives the same leading approximation from the factorial expression for $\binom{n}{m}$.

Summary

- Complex differentiability requires one direction-independent difference-quotient limit. With continuous first partial derivatives, this is equivalent to the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$.
- Cauchy's theorem implies path independence on simply connected domains. Cauchy's integral formula yields all derivatives and the local Taylor expansion.
- A Laurent series is valid on an annulus. Its principal part classifies an isolated singularity as removable, a pole, or essential. Branch points are not isolated and do not possess ordinary residues.
- The residue is the coefficient a_{-1} . The residue theorem evaluates a closed contour integral as $2\pi i$ times the sum of enclosed residues.
- Three real-integral templates: semicircle (rational/exponential), keyhole (powers and branch cuts, as in $\int_0^\infty x^a/(1+x)^2 dx$), and Jordan's indented semicircle (oscillatory integrals such as $\int \sin x/x dx$). Always check orientation and vanishing arcs before taking limits.
- **Steepest descent** (1.9.4) is a local Gaussian approximation at a nondegenerate dominant saddle. Real special cases: Laplace method (1.9.5) (Stirling in Ch. 2) and stationary phase (1.9.6) (large- x Bessel in Ch. 5). Orientation and the phase of the square root are part of the answer.